

# Foundations of Fixed Income and Derivative Instruments

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# 1 The Time Value of Money

## 1.1 Why a Euro Today Is Not a Euro Tomorrow

Pricing formulas look complicated, but they rest on three simple ideas. First: identify the cash flows that will change hands in the future. Second: compute what those cash flows are worth today. Third: if two strategies deliver the same future cash, they must cost the same today — otherwise there is an arbitrage.

1. Identify future cash flows.
2. Compute what they are worth today (present value).
3. If two strategies lead to the same cash flows, they must cost the same (no-arbitrage).

The rest of this chapter develops the language of interest rates that makes those three steps precise.

## 1.2 Discrete Compounding

This section builds from a one-year deposit to the general formula with  $m$  compounding periods per year. Each subsection starts with numbers and then states the corresponding formula.

### 1.2.1 One year, one compound

Take  $PV = 1000$  euros, quoted annual rate  $r = 5\%$ , one year, interest paid once.

$\text{Interest} = PV \times r = 1000 \times 0.05 = 50$ .

$$FV = PV + PV \times r = PV (1 + r) = 1000 \times 1.05 = 1050$$

There is no compounding frequency yet: one year, one interest payment.

#### Python — one year, one compound

```
PV = 1000
r = 0.05
interest = PV * r
FV = PV * (1 + r)
print(f"interest = {interest}, FV = {FV}")
```

### 1.2.2 Annual compounding over several years

The same contract, repeated. After year 1 the balance is 1050. Year 2 applies 5% to 1050, not to 1000:

$$1050 \times 1.05 = 1102.50$$

Year 3:

$$1102.50 \times 1.05 = 1157.625 \approx 1157.63$$

The extra beyond  $3 \times 50$  is interest on interest — compounding. The pattern is:

$$FV = PV (1 + r)^T$$

with  $T = 3$ ,  $r = 5\%$ , annual compounding.

#### Python — annual compounding, three years

```
PV, r, T = 1000, 0.05, 3
year1 = PV * (1 + r)
year2 = year1 * (1 + r)
year3 = year2 * (1 + r)
FV = PV * (1 + r) ** T
print(f"year1 = {year1}, year2 = {year2}, year3 = {year3}, FV = {FV}")
```

### 1.2.3 Semiannual compounding and the role of $m$

Keep the same *quoted* annual rate  $r = 5\%$ , but pay interest twice a year. Each half-year the rate is  $r/2 = 2.5\%$ . Over one year:

$$FV = PV (1 + r/2)^2 = 1000 \times (1.025)^2 = 1050.625$$

Compared with annual compounding over one year (1050), the same quoted 5% produces slightly more money when compounding is more frequent.

Let  $m$  be the number of compounding periods per year. The general discrete formula is:

$$FV = PV (1 + r/m)^{mT}$$

Annual compounding is  $m = 1$ ; semiannual is  $m = 2$ ; monthly is  $m = 12$ .

#### Python — semiannual vs annual, one year

```
PV, r, T = 1000, 0.05, 1
FV_annual = PV * (1 + r) ** T
m = 2
FV_semiannual = PV * (1 + r / m) ** (m * T)
print(f"FV_annual = {FV_annual}, FV_semiannual = {FV_semiannual}")
```

**Key Insight.** When the *quoted* rate is fixed, more frequent compounding produces a larger future value. That statement is different from comparing two quoted rates that describe the *same* cash flows — the situation treated in the worked examples below.

### 1.3 Continuous Compounding and the Number $e$

Fix the quoted rate at  $r = 10\%$ ,  $PV = 1000$ ,  $T = 1$ , and increase  $m$ :

| $m$      | Convention | FV         |
|----------|------------|------------|
| 1        | Annual     | \$1,100.00 |
| 2        | Semiannual | \$1,102.50 |
| 12       | Monthly    | \$1,104.71 |
| 365      | Daily      | \$1,105.16 |
| $\infty$ | Continuous | \$1,105.17 |

As  $m \rightarrow \infty$ , growth becomes continuous:

$$FV = PV e^{rT}$$

Continuous compounding is the same growth applied in infinitely small steps — that is why  $e$  appears throughout derivatives pricing.

**Python — fixed quote 10%, vary m**

```
import numpy as np

PV, r, T = 1000, 0.10, 1
for m in (1, 2, 12, 365):
    FV = PV * (1 + r / m) ** (m * T)
    print(f"m = {m}, FV = {FV}")
print(f"continuous FV = {PV * np.exp(r * T)}")
print(f"EAR semiannual = {(1 + r / 2) ** 2 - 1}")
```

### 1.4 Present Value and Discounting

Discounting reverses compounding. Given a future cash flow FV at time  $T$ :

$$PV = \frac{FV}{(1 + r)^T} \quad (\text{annual})$$

$$PV = \frac{FV}{(1 + r/m)^{mT}} \quad (\text{m times per year})$$

$$PV = FV e^{-rT} \quad (\text{continuous})$$

**Key Insight.** A euro today is worth more than a euro tomorrow because today's euro can be invested and earn interest. Discounting answers: how much must be invested today at rate  $r$  to obtain FV in  $T$  years?

## 1.5 Effective Annual Rate

When interest is compounded more than once per year, the *effective annual rate* (EAR) exceeds the quoted (nominal) rate. With  $m$  periods per year at nominal rate  $r$ :

$$\text{EAR} = (1 + r/m)^m - 1$$

Under continuous compounding:  $\text{EAR} = e^r - 1$ .

### Semiannual compounding at a 10% quote

At a nominal rate of 10% with semiannual compounding ( $m = 2$ ):

$$\text{EAR} = (1 + 0.10/2)^2 - 1 = (1.05)^2 - 1 = 0.1025 = 10.25\%$$

The effective rate is 10.25%, not 10%. Mid-year interest itself earns interest in the second half of the year.

## 1.6 Same Cash, Four Quoted Rates

The following worked example is Practice Question 4.3 from Hull (2022).<sup>1</sup> Suppose \$1,000 today grows to \$1,100 in one year — the *same* cash under every quoting convention. The task is to recover the quoted rate  $r$  implied by each convention.

### 1.6.1 (a) Annual compounding

Using  $\text{FV} = \text{PV} (1 + r)^T$  with  $T = 1$ :

$$1100 = 1000(1 + r)$$

$$1100/1000 = 1 + r$$

$$1.1 = 1 + r$$

$$r = 1.1 - 1 = 0.10 = 10\%$$

(Equivalently  $r = \text{FV}/\text{PV} - 1$ .)

#### Python — annual quote from 1000 to 1100

```
PV, FV = 1000, 1100
r = FV / PV - 1
print(f"r = {r}, r (%) = {r * 100}")
```

### 1.6.2 (b) Semiannual compounding ( $m = 2$ )

Using  $\text{FV} = \text{PV} (1 + r/2)^2$ :

$$1100 = 1000(1 + r/2)^2$$

<sup>1</sup>Hull, J. C. (2022). *Options, futures, and other derivatives* (11th ed., Global). Pearson. Practice Question 4.3.

$$1.1 = (1 + r/2)^2$$

$$\sqrt{1.1} = 1 + r/2$$

$$\sqrt{1.1} - 1 = r/2$$

$$r = 2(\sqrt{1.1} - 1) \approx 0.097618 = 9.76\%$$

**Python — semiannual quote from 1000 to 1100**

```
PV, FV, m = 1000, 1100, 2
r = m * ((FV / PV) ** (1 / m) - 1)
print(f"r = {r}, r (%) = {r * 100}")
```

### 1.6.3 (c) Monthly compounding ( $m = 12$ )

Using  $FV = PV (1 + r/12)^{12}$ :

$$1100 = 1000(1 + r/12)^{12}$$

$$1.1 = (1 + r/12)^{12}$$

$$1.1^{1/12} = 1 + r/12$$

$$1.1^{1/12} - 1 = r/12$$

$$r = 12(1.1^{1/12} - 1) \approx 0.095690 = 9.57\%$$

The same steps work for any  $m$  when  $T = 1$ . Start from the discrete growth law:

$$FV = PV (1 + r/m)^m$$

$$FV / PV = (1 + r/m)^m$$

$$(FV / PV)^{1/m} = 1 + r/m$$

$$(FV / PV)^{1/m} - 1 = r/m$$

$$r = m((FV / PV)^{1/m} - 1)$$

**Python — monthly quote from 1000 to 1100**

```
PV, FV, m = 1000, 1100, 12
r = m * ((FV / PV) ** (1 / m) - 1)
print(f"r = {r}, r (%) = {r * 100}")
```

### 1.6.4 (d) Continuous compounding

Using  $FV = PV e^{rT}$  with  $T = 1$ :

$$1100 = 1000e^r$$

$$1.1 = e^r$$

$$\ln(1.1) = r$$

$$r \approx 0.095310 = 9.53\%$$

(Equivalently  $r = \ln(\text{FV} / \text{PV})$  when  $T = 1$ .)

#### Python — continuous quote from 1000 to 1100

```
import numpy as np

PV, FV = 1000, 1100
r = np.log(FV / PV)
print(f"r = {r}, r (%) = {r * 100}")
```

**Key Insight.** The same cash flows imply four different quoted rates. For a *fixed* payoff, the quoted rate *falls* as compounding becomes more frequent. More frequent compounding raises the future value when the *quote* is held fixed; it does not raise the quote when the cash flows are held fixed.

## 1.7 Same Method Over Five Years

The same inversion, with a five-year horizon: recover the quoted rate that turns money today into money later under annual, semiannual, monthly, and continuous compounding.

PV = 2000, FV = 3600,  $T = 5$  years.

### 1.7.1 (a) Annual compounding

Using  $\text{FV} = \text{PV} (1 + r)^T$ :

$$3600 = 2000(1 + r)^5$$

$$3600/2000 = (1 + r)^5$$

$$1.8 = (1 + r)^5$$

$$1.8^{1/5} = 1 + r$$

$$r = 1.8^{1/5} - 1 \approx 0.1247 = 12.47\%$$

#### Python — annual quote, five years

```
PV, FV, T = 2000, 3600, 5
r = (FV / PV) ** (1 / T) - 1
print(f"r = {r}, r (%) = {r * 100}")
```

### 1.7.2 (b) Semiannual compounding ( $m = 2$ )

Using  $FV = PV (1 + r/2)^{2T} = PV (1 + r/2)^{10}$ :

$$3600 = 2000(1 + r/2)^{10}$$

$$1.8 = (1 + r/2)^{10}$$

$$1.8^{1/10} = 1 + r/2$$

$$1.8^{1/10} - 1 = r/2$$

$$r = 2(1.8^{1/10} - 1) \approx 0.1211 = 12.11\%$$

**Python — semiannual quote, five years**

```
PV, FV, T, m = 2000, 3600, 5, 2
r = m * ((FV / PV) ** (1 / (m * T)) - 1)
print(f"r = {r}, r (%) = {r * 100}")
```

### 1.7.3 (c) Monthly compounding ( $m = 12$ )

Using  $FV = PV (1 + r/12)^{12T} = PV (1 + r/12)^{60}$ :

$$3600 = 2000(1 + r/12)^{60}$$

$$1.8 = (1 + r/12)^{60}$$

$$1.8^{1/60} = 1 + r/12$$

$$1.8^{1/60} - 1 = r/12$$

$$r = 12(1.8^{1/60} - 1) \approx 0.1181 = 11.81\%$$

For a general horizon  $T$ , start again from the discrete growth law:

$$FV = PV (1 + r/m)^{mT}$$

$$FV / PV = (1 + r/m)^{mT}$$

$$(FV / PV)^{1/(mT)} = 1 + r/m$$

$$(FV / PV)^{1/(mT)} - 1 = r/m$$

$$r = m((FV / PV)^{1/(mT)} - 1)$$

**Python — monthly quote, five years**

```
PV, FV, T, m = 2000, 3600, 5, 12
r = m * ((FV / PV) ** (1 / (m * T)) - 1)
print(f"r = {r}, r (%) = {r * 100}")
```

### 1.7.4 (d) Continuous compounding

Using  $FV = PV e^{rT}$ :

$$3600 = 2000e^{5r}$$

$$1.8 = e^{5r}$$

$$\ln(1.8) = 5r$$

$$r = \ln(1.8)/5 \approx 0.1176 = 11.76\%$$

(Equivalently  $r = \ln(FV / PV)/T$ .)

**Python — continuous quote, five years**

```
import numpy as np

PV, FV, T = 2000, 3600, 5
r = np.log(FV / PV) / T
print(f"r = {r}, r (%) = {r * 100}")
```

Again the quoted rate falls as compounding becomes more frequent, for the same cash over five years: about 12.47%, 12.11%, 11.81%, and 11.76%.

## 1.8 A Forward Is Not a Forecast: Gold Carry Check

A forward (or futures) price is often mistaken for a prediction of where the spot will be at maturity. It is not. It is a *financing* price: the level at which borrowing to buy the spot and locking delivery through the forward leaves no easy free lunch from interest rates alone.

### 1.8.1 Interest-only approximation

Ignore storage, lease income, day-count nuance, and spot-versus-futures differences for a moment. With spot  $S_0$ , financing rate  $r$ , and time  $T$  to the forward's maturity, the interest-only fair forward is the annual carry already derived:

$$F_{\text{fair},r} = S_0(1 + r)^T$$

Compare that to the market quote  $F_{\text{mkt}}$ . The residual

$$F_{\text{mkt}} - F_{\text{fair},r}$$

is not automatically arbitrage. On a liquid metal it usually reflects:

- **Storage** — vaulting and insurance raise the cost of holding physical gold, which pushes the forward *up* versus pure interest carry.
- **Gold lease rate** — income from lending gold (for example to a jeweller or a short seller) is a benefit of holding the spot asset, like a dividend; that pushes the forward *down*.
- **Convenience yield** — the non-cash benefit of having the physical metal on hand rather than only a futures claim; the lease rate is the market's observable cousin of that idea for gold.

- Day-count conventions and the difference between OTC forwards and exchange futures.

### Market illustration (7 September 2026)

Bloomberg mid quotes on that date (an illustration, not a permanent market level):

- Gold spot  $S_0 = 4394$  USD/oz (XAU)
- Near one-year forward proxy  $F_{\text{mkt}} = 4620.3$  USD/oz (COMEX gold future; expiry 27 August 2027,  $T \approx 0.97$ )
- USD financing  $r = 4.1398\%$  (1Y SOFR OIS fixed mid)

$$F_{\text{fair},r} = 4394 \times (1.041398)^{0.97} \approx 4570.34$$

|                                               |            |
|-----------------------------------------------|------------|
| $F_{\text{fair},r}$                           | \$4,570.34 |
| $F_{\text{mkt}}$                              | \$4,620.30 |
| residual $F_{\text{mkt}} - F_{\text{fair},r}$ | \$49.96    |

About \$50/oz sits in storage / lease / day-count / instrument mismatch — a carry check, not a free lunch. The financing quote used here is not continuously compounded; the annual form  $S_0(1+r)^T$  is the appropriate approximation (see the next section).

#### Python — gold carry check

```
# 1Y SOFR OIS fixed is annual Act/360, not continuous.
S0 = 4394.0
F_mkt = 4620.3
r = 0.041398
T = 0.97

F_fair_r = S0 * (1 + r) ** T
residual = F_mkt - F_fair_r
print(f"F_fair_r = {F_fair_r}")
print(f"F_mkt = {F_mkt}")
print(f"residual = {residual}")
```

**Key Insight.** The forward price is a financing price, not a forecast of the future spot. The same three ideas — identify cash flows, move them through time, and rule out free lunches — reappear for forwards and options in later chapters.

## 1.9 The Financing Rate in the Gold Example

The rate  $r = 4.1398\%$  above is the fixed rate on a one-year *SOFR OIS* (overnight index swap): annual fixed payments, Act/360 — not continuous compounding. Overnight SOFR alone is the wrong input for a one-year financing check.

**SOFR** (Secured Overnight Financing Rate) is a near risk-free USD overnight rate published by the New York Fed from actual Treasury-collateralised repo transactions. The loan is for one business day; collateral keeps credit risk low relative to unsecured bank-to-bank rates.

**Act/360:** interest for a period uses the actual number of calendar days and a 360-day year,

$$\text{Interest} \approx \text{Principal} \times r \times (\text{days}/360).$$

This chapter uses  $S_0(1+r)^T$  as a rough carry check; it does not rebuild full OIS / Act/360 pricing.

Before SOFR, many USD contracts referenced LIBOR — an unsecured panel survey rate. USD panel LIBOR was wound down in 2023. When older textbooks still write “LIBOR,” the floating-versus-fixed mechanics are unchanged, but modern markets refer to SOFR / OIS.

## 1.10 Chapter Summary and Formula Sheet

**Future value (discrete):**  $FV = PV (1 + r/m)^{mT}$

**Future value (continuous):**  $FV = PV e^{rT}$

**Present value (annual / with  $m$  / continuous):**

$$PV = FV / (1 + r)^T, \quad PV = FV / (1 + r/m)^{mT}, \quad PV = FV e^{-rT}$$

**Effective annual rate:**  $EAR = (1 + r/m)^m - 1, \quad EAR_{\text{cont}} = e^r - 1$

**Invert discrete quote ( $T = 1$ ):**  $r = m((FV / PV)^{1/m} - 1)$

**Invert discrete quote (general  $T$ ):**  $r = m((FV / PV)^{1/(mT)} - 1)$

**Invert continuous quote:**  $r = \ln(FV / PV)/T$

**Interest-only forward carry (approximation):**  $F_{\text{fair},r} \approx S_0(1+r)^T$

## 2 Forward Contracts

### 2.1 Why Agree on a Price Today for a Trade Tomorrow

Chapter 1 turned cash through time with interest rates. This chapter adds a second costume for the same three ideas: a contract that locks a price *now* for an exchange that happens *later*.

1. Identify the future exchange (what is bought or sold, how much, and when).
2. Value the position by comparing the locked delivery price to the market at maturity  $T$ .
3. If two routes deliver the same future cash, they must cost the same today — otherwise someone can lock a risk-free profit (an *arbitrage*).

A *forward* is the simplest such contract. The rest of the chapter builds long and short P&L, the forward price  $F_0$ , and the cash-and-carry trade when the quoted forward is wrong.

### 2.2 Spot Versus Forward

A *spot* trade settles essentially now: cash and the asset change hands today (or on the market's short spot settlement cycle).

A *forward* trade agrees *today* on the terms, but the exchange of asset against cash happens on the maturity date  $T$ . At inception, the usual market convention is to choose the delivery price so that the forward itself is worth zero to enter — no upfront payment to go long or short.

#### Forward Contract

A binding agreement between two parties to buy or sell an underlying asset at a predetermined delivery price  $K$  on a specified maturity date  $T$ .



Figure 1: Spot settles essentially at  $t = 0$ . A forward locks terms at  $t = 0$  and exchanges at maturity  $T$ .

Two prices must not be confused:

- The delivery price  $K$  is locked in the contract once the deal is struck. It does not change.
- $F_0$  (or  $F_0(T)$ ) is the *forward price*: the delivery price that would make a *brand-new* forward for maturity  $T$  worth zero today. As time and markets move,  $F_0$  can change even though an old contract's delivery price  $K$  stays fixed.

When a new forward is struck at mid, the counterparties set the delivery price  $K = F_0$ . Afterward, if the live forward price moves away from that delivery price  $K$ , the old contract acquires value (positive for one side, negative for the other).

## 2.3 Long Versus Short

Every forward has two sides.

- **Long forward:** obligation to *buy* the asset at the delivery price  $K$  at maturity  $T$ . Gains if the spot at maturity  $T$ ,  $S_T$ , is above the delivery price  $K$ .
- **Short forward:** obligation to *sell* the asset at the delivery price  $K$  at maturity  $T$ . Gains if  $S_T$  is below the delivery price  $K$ .

One side's gain is the other's loss: settlement P&L is zero-sum.

**Key Insight.** Spot means trade now. Forward means agree now and exchange at maturity  $T$ . The long is committed to buy at the delivery price  $K$ ; the short is committed to sell at the delivery price  $K$ .

## 2.4 Why Enter a Forward: Hedge or Speculate

A forward is optional in the economic sense: nobody is forced to lock a price. Two common motives explain most of the volume students meet first.

### 2.4.1 Hedging

A *hedge* uses the forward to reduce a risk that already exists in the real business.

**Exporter with foreign-currency receivables.** A US firm will receive 100,000 GBP in three months for services already sold. If GBP depreciates against USD, the same pounds buy fewer dollars. Selling GBP forward (a *short* forward on GBP, quoted as USD per GBP) locks a dollar value for those receivables.

**Commodity producer.** A wheat farmer harvests later in the year. If the wheat price falls, revenue falls even though the work is done. Selling wheat forward locks a sale price.

**Commodity consumer.** A cereal manufacturer must buy wheat continuously and cannot freely pass every spike into supermarket prices. Buying wheat forward (a *long* forward) locks a purchase price.

**Airline fuel.** Jet fuel is a large, volatile cost. Airlines often lock part of their future fuel bill with forwards or closely related contracts so that a spike in energy prices does not wipe out the ticket margin.

In each case the forward does not invent exposure: it *offsets* an exposure the firm already has.

### 2.4.2 Speculating

A *speculation* uses the forward to take a view with *no* matching inventory or receivable.

If you are long the forward with no need for the asset, you are betting that the spot at maturity  $T$  will finish above the delivery price  $K$ . If you are short with nothing to sell, you are betting it finishes below. There is no “underlying” cash-flow hedge — only P&L on the contract.

### 2.4.3 A third cast member: arbitrage

A third motive appears when related prices are inconsistent — for example when a quoted forward does not line up with the cost of financing the spot asset. Locking a risk-free profit from that inconsistency is *arbitrage*. We return to it when we price  $F_0$  and when we work the gold cash-and-carry.

**Key Insight.** Hedge: reduce a pre-existing business risk. Speculate: take a naked view on where the spot will be at maturity  $T$  relative to the delivery price  $K$ . Arbitrage: exploit inconsistent prices; details later in the chapter.

## 2.5 Worked Example: FX Forward P&L

An investor enters a short forward to sell 100,000 GBP for USD at a fixed delivery price  $K$ . What is the P&L if, at maturity  $T$ , the spot (USD per GBP) is (a) one cent below the delivery price  $K$ , or (b) two cents above it?<sup>2</sup>

### Market illustration (9 September 2026)

On that date the short-dated GBPUSD forward outright was

$$F_0 = 1.35465$$

USD per GBP

Take this as the delivery price of a newly struck short-dated forward:  $K = F_0 = 1.35465$ .

Notional 100,000 GBP. Scenario spots at maturity  $T$ :

- (a)  $S_T = 1.34465$  (one cent below the delivery price  $K$ )
- (b)  $S_T = 1.37465$  (two cents above the delivery price  $K$ )

### 2.5.1 Case (a): $S_T = 1.34465$

The short is committed to *sell* GBP at the delivery price  $K = 1.35465$  while the market is at 1.34465. Selling above the market is profitable:

$$\text{P\&L}_{\text{short}} = 100000 \times (1.35465 - 1.34465) = 100000 \times 0.01 = +1000$$

The long is the other side of the same contract and therefore loses \$1,000.

**Python — FX P&L, case (a)**

<sup>2</sup>Hull, J. C. (2022). *Options, futures, and other derivatives* (11th ed., Global). Pearson. Practice Question 1.2; numbers updated.

```

quantity = 100_000
K = 1.35465
S_T = 1.34465
pnl_short = quantity * (K - S_T) # short gains when S_T < K
pnl_long = quantity * (S_T - K)
print(f"pnl_short = {pnl_short}, pnl_long = {pnl_long}")

```

### 2.5.2 Case (b): $S_T = 1.37465$

Now the market is above the delivery price  $K$ . The short must sell GBP cheap relative to the spot and loses:

$$\text{P\&L}_{\text{short}} = 100000 \times (1.35465 - 1.37465) = 100000 \times (-0.02) = -2000$$

The long gains \$2,000.

#### Python — FX P&L, case (b)

```

quantity = 100_000
K = 1.35465
S_T = 1.37465
pnl_short = quantity * (K - S_T)
pnl_long = quantity * (S_T - K)
print(f"pnl_short = {pnl_short}, pnl_long = {pnl_long}")

```

## 2.6 The General P&L Formula

The two cases suggest a single expression. Define

Position = +1 for a long forward

Position = -1 for a short forward

Then, with quantity measured in units of the underlying,

$$\text{P\&L} = \text{Quantity} \times (S_T - K) \times \text{Position}$$

For an FX forward quoted as USD per GBP, a GBP notional makes the P&L come out in USD. Check against the example: in case (a),  $S_T - K = -0.01$ , so the short (Position = -1) earns  $100000 \times (-0.01) \times (-1) = +1000$ .

#### Python — FX P&L with Position

```

def pnl(quantity, S_T, K, position):
    return quantity * (S_T - K) * position

quantity = 100_000
K = 1.35465

```

```
# Case (a)
S_T = 1.34465
print("a short", pnl(quantity, S_T, K, -1))
print("a long ", pnl(quantity, S_T, K, +1))

# Case (b)
S_T = 1.37465
print("b short", pnl(quantity, S_T, K, -1))
print("b long ", pnl(quantity, S_T, K, +1))
```

**Key Insight.** Compute  $(S_T - K)$  once, then multiply by Position. The short P&L is exactly minus the long P&L: settlement is zero-sum.

## 2.7 The Fair Forward Price (No Income)

Settlement P&L compares  $S_T$  to a *locked* delivery price  $K$ . Which delivery price should a *new* forward use so that neither side pays to enter? That special choice is the *forward price*  $F_0$ .

### Forward Price $\$F_0$

The delivery price  $K$  that makes a newly initiated forward for maturity  $T$  worth zero today. Once the deal is struck at that level, the contract's delivery price stays fixed at  $K = F_0$  even if the live forward price later moves.

Consider a stock that pays no dividend (and ignore storage). Two strategies both deliver one share at maturity  $T$ :

1. **Financed spot.** Borrow  $S_0$  today, buy the share, and hold it to maturity  $T$ . At maturity  $T$  you own the share and owe the financing repayment.
2. **Long forward.** Enter a long forward with delivery price  $K$ . No cash changes hands today. At maturity  $T$  you pay  $K$  and receive the share.

Same terminal share, different cash at maturity  $T$ . If one route is systematically cheaper, an arbitrageur can lock the difference. No-arbitrage therefore forces the forward's delivery price to equal the financing cost of the spot route. That common level is  $F_0$ .

### 2.7.1 Continuous compounding

If the financing rate  $r$  is quoted with continuous compounding,

$$F_0 = S_0 e^{rT}.$$

**Six-month forward on a non-dividend stock**

A 6-month forward on a non-dividend-paying stock:  $S_0 = 30$ ,  $r = 5\%$  per annum with continuous compounding,  $T = 0.5$ .<sup>3</sup>

$$F_0 = 30e^{0.05 \times 0.5} = 30e^{0.025} \approx 30.7595.$$

So  $F_0 \approx \$30.76$ .

**Python — fair F0, continuous**

```
import math
S0 = 30.0
r = 0.05
T = 0.5
F0 = S0 * math.exp(r * T)
print(f"F0 = {F0}")
```

**2.7.2 Annual discrete compounding**

The same quoted rate  $r = 5\%$ , read instead as an annually compounded rate, gives a different fair forward:

$$F_0 = S_0(1 + r)^T = 30(1.05)^{0.5} \approx 30.7409.$$

**Python — fair F0, annual discrete**

```
S0 = 30.0
r = 0.05
T = 0.5
F0 = S0 * (1 + r) ** T
print(f"F0 = {F0}")
```

Same  $S_0$ , same printed  $r$ , same maturity  $T$  — two compounding dialects, two fair forwards. Textbooks often work in continuous compounding because it makes later derivative-pricing mathematics more convenient, for reasons that appear in later chapters. The street quotes discrete rates. Strictly speaking, only discrete compounding exists in the market: no asset pays interest at every instant, so continuous compounding is an idealisation that approximates very frequent compounding rather than a literal cash-flow schedule.

**Key Insight.**  $F_0$  is a financing price for delivery at maturity  $T$ , not a forecast of  $S_T$ . Continuous and discrete dialects that print the same  $r$  produce slightly different fair forwards; always match the formula to the quote.

<sup>3</sup>Hull, J. C. (2022). *Options, futures, and other derivatives* (11th ed., Global). Pearson. Practice Question 5.2.

## 2.8 When the Quoted Forward Is Wrong: Cash-and-Carry

### 2.8.1 Cash-and-carry

#### Market illustration (9 September 2026)

Bloomberg mids that day for a near one-year gold check:

- Spot  $S_0 = 4401.56$  USD/oz (XAU Curncy)
- Forward proxy  $F_{\text{mkt}} = 4588.15$  USD/oz (GCQ7 Comdty; last tradeable 27 August 2027,  $T \approx 0.9637$ )
- Financing  $r = 4.1723\%$  (US05FR1 Curncy — 1Y SOFR OIS fixed)

For the mechanics of cash-and-carry, set storage and lease income to zero and treat the futures mid as an OTC forward quote. Those shortcuts are for illustration only: in a live gold market they have to be priced in (storage and lease return below).

**Today.** Borrow \$4,401.56 at 4.1723%; buy one ounce in the spot market; short the forward at the delivery price  $K = 4588.15$ .

**At maturity  $T$ .** Deliver the ounce under the forward and receive \$4,588.15; repay the loan, which has grown to

$$4401.56(1.041723)^{0.9637} \approx 4578.40.$$

Under that interest-only story the residual looks lockable:

$$\text{Locked profit} = 4588.15 - 4578.40 \approx 9.75$$

#### Python — cash-and-carry lock

```
S0 = 4401.56
F = 4588.15
r = 0.041723
T = 0.9637
F0 = S0 * (1 + r) ** T
profit = F - F0
print(f"F0 = {F0}")
print(f"locked profit = {profit}")
```

### 2.8.2 Three terminal spots, one locked profit

The cash-and-carry trade above looks suspiciously neat: borrow, buy the ounce, short the forward, and pocket about \$9.75 at maturity  $T$ . A natural worry is that the ending spot price could spoil the story. What if gold rallies far above the delivery price  $K$ ? What if it collapses? What if it finishes exactly at  $K$ ?

The answer is that *none of those movements changes the locked profit*. The short forward is there precisely so that a move in the spot is offset by an opposite move on the forward. The trade is a financing lock, not a forecast of where gold will be at maturity  $T$ .

There are two equivalent ways to see what happens at maturity  $T$ :

1. **Physical delivery under the forward.** Hand over the ounce you already hold and receive the delivery price  $K$ . You never sell in the spot market, so  $S_T$  never appears in the cash flows. You repay the loan and keep  $K$  minus the loan repayment.
2. **Spot sale plus cash settlement of the short.** Sell the ounce in the spot market for  $S_T$ . The short forward then settles for  $K - S_T$ : you gain when  $S_T$  is below the delivery price  $K$ , and you pay when  $S_T$  is above it. Adding the two pieces gives

$$S_T + (K - S_T) = K.$$

The  $S_T$  terms cancel. You are back to receiving the delivery price  $K$  in net, exactly as under physical delivery. After you repay the loan, the same residual remains.

So the only comparison that matters is the delivery price  $K$  against the financed cost of holding the ounce — not the path of the spot between today and maturity  $T$ .

Line up the cash flows on a timeline. Today every line that puts cash in your hand is matched by a line that takes it out; the forward itself is free to enter. All of the locked profit sits at maturity  $T$ .

**Today ( $t = 0$ ) — same for every later ending of  $S_T$ :**

| Line                                              | Cash (USD) |
|---------------------------------------------------|------------|
| Borrow $S_0$ at 4.1723%                           | +4401.56   |
| Buy one ounce in the spot market                  | −4401.56   |
| Short the forward at delivery price $K = 4588.15$ | 0          |
| <b>Net today</b>                                  | <b>0</b>   |

You hold one ounce, you owe the loan, and you are short the forward. No cash has left your pocket yet.

**Maturity  $T$  — physical delivery under the forward ( $S_T$  never appears):**

| Line                                              | Cash (USD)   |
|---------------------------------------------------|--------------|
| Deliver the ounce; receive the delivery price $K$ | +4588.15     |
| Repay the loan $S_0(1 + r)^T$                     | −4578.40     |
| <b>Net at maturity <math>T</math></b>             | <b>+9.75</b> |

That is the cleanest picture of the lock. The next example opens the same maturity  $T$  into a spot sale plus short settlement, so you can watch  $S_T$  appear and then cancel.

#### Same cash-and-carry, three values of $S_T$

Keep the trade from the market illustration: delivery price  $K = 4588.15$ , loan due  $\approx 4578.40$ . Today is unchanged — net cash at  $t = 0$  is still zero. At maturity  $T$ , use the second view — sell the ounce at  $S_T$  and settle the short forward for  $K - S_T$  — and try three deliberately different endings.

**Case  $S_T > K$ .** Suppose gold finishes at  $S_T = 4700$ . Selling the ounce brings in \$4,700, which looks wonderful on its own. But the short forward is underwater: you owe

$$K - S_T = 4588.15 - 4700 = -111.85.$$

Spot sale plus short settlement:  $4700 - 111.85 = 4588.15$ . After the loan, net  $\approx 9.75$ . The rally enriched the metal you hold and hurt the short by the same amount.

**Case  $S_T = K$ .** Suppose gold finishes exactly at the delivery price,  $S_T = 4588.15$ . The spot sale brings in \$4,588.15 and the short forward settles at zero — there is nothing to gain or lose against a delivery price that matches the spot. After the loan, the net is again  $\approx 9.75$ . This middle case makes the cancellation obvious: when  $S_T = K$ , the forward is idle and the whole residual is just  $K$  versus the loan.

**Case  $S_T < K$ .** Suppose gold finishes at  $S_T = 4400$ . The ounce sells for only \$4,400 — a painful print if you were unhedged. The short forward more than offsets that pain:

$$K - S_T = 4588.15 - 4400 = 188.15.$$

Spot sale plus short settlement:  $4400 + 188.15 = 4588.15$ . After the loan, net  $\approx 9.75$  once more. The drop hurt the metal you hold and paid you on the short by the same amount.

The same maturity- $T$  cash flows in one grid. Each column is one ending of  $S_T$ ; every column still nets to about \$9.75 after the loan:

| Line at maturity $T$       | $S_T > K$    | $S_T = K$    | $S_T < K$    |
|----------------------------|--------------|--------------|--------------|
| $S_T$                      | 4700.00      | 4588.15      | 4400.00      |
| Sell the ounce ( $+S_T$ )  | +4700.00     | +4588.15     | +4400.00     |
| Short settle ( $K - S_T$ ) | -111.85      | 0.00         | +188.15      |
| Subtotal ( $= K$ )         | +4588.15     | +4588.15     | +4588.15     |
| Repay the loan             | -4578.40     | -4578.40     | -4578.40     |
| <b>Net</b>                 | <b>+9.75</b> | <b>+9.75</b> | <b>+9.75</b> |

In every column, the spot sale plus the short settlement equals the delivery price  $K = 4588.15$  — the same subtotal you get by delivering the ounce under the forward. Subtract the loan:

$$4588.15 - 4578.40 \approx 9.75.$$

That \$9.75 was fixed *today*, when the quoted forward sat above the financed cost of the ounce. Nothing that happens to  $S_T$  afterward opens or closes the gap. Today was always a wash; maturity  $T$  is where the lock is collected.

#### Python — lock vs three terminal spots

```
K = 4588.15
loan = 4578.40
for label, S_T in (
    ("above K", 4700.00),
```

```

("equal K", 4588.15),
("below K", 4400.00),
):
    spot_sale = S_T
    short_settle = K - S_T
    subtotal = spot_sale + short_settle
    net = subtotal - loan
    print(label, spot_sale, short_settle, subtotal, net)

```

**Key Insight.** Cash-and-carry is not a bet on whether gold will rise or fall. Today ( $t = 0$ ) the cash flows net to zero. At maturity  $T$ , the short forward cancels the spot move, whether  $S_T$  finishes above, at, or below the delivery price  $K$ . The locked profit is only the gap between that delivery price  $K$  and the financed cost of holding the ounce.

The loan repayment  $\approx 4578.40$  is the financed cost of holding the ounce to maturity  $T$ . Under zero storage and zero income, that cost is the annual no-arbitrage forward

$$F_0 = S_0(1 + r)^T.$$

If a quoted forward for the same maturity  $T$  sits away from  $F_0$ , the two routes to the same future delivery no longer cost the same, and a locked trade can harvest the gap under those assumptions.

- If the quoted forward is *above*  $F_0$ , the forward is too rich: borrow cash, buy the spot, and *short* the forward (*cash-and-carry* — the trade just shown).
- If the quoted forward is *below*  $F_0$ , the forward is too cheap: sell the spot, lend the proceeds, and *long* the forward (*reverse cash-and-carry*).

### 2.8.3 Reverse cash-and-carry

#### Gold forward below financed spot

Keep the same spot, financing rate, and maturity  $T$  as above, so the financed cost is again  $F_0 \approx 4578.40$ , but now suppose the quoted forward were only  $F = 4400$ .

**Today.** Sell one ounce in the spot market for \$4,401.56; lend the proceeds at 4.1723%; long the forward at the delivery price  $K = 4400$ .

**At maturity  $T$ .** The loan repays  $\approx 4578.40$ . Pay the delivery price  $K = 4400$  on the long forward, receive the ounce, and replace what was sold.

$$\text{Locked profit} = 4578.40 - 4400 \approx 178.40$$

Python — reverse cash-and-carry lock

```

S0 = 4401.56
F = 4400.0
r = 0.041723
T = 0.9637
F0 = S0 * (1 + r) ** T
profit = F0 - F
print(f"F0 = {F0}")
print(f"locked profit = {profit}")

```

### 2.8.4 Three terminal spots, one locked profit (reverse)

Reverse cash-and-carry is the mirror image. The quoted forward is *below* the financed cost of the ounce, so you sell the spot today, lend the cash, and go *long* the forward at the cheap delivery price  $K$ . The same worry appears: what if gold rallies or collapses before maturity  $T$ ?

Again, *none of those endings changes the lock*. The long forward is there so that whatever you pay to buy the ounce back in the spot market is offset by the forward settlement. You are not betting on  $S_T$ ; you are harvesting the gap between the loan repayment and the delivery price  $K$ .

Two equivalent views at maturity  $T$ :

1. **Physical delivery under the long forward.** Collect the loan repayment, pay the delivery price  $K$ , receive the ounce back — replacing what you sold at  $t = 0$ . The spot price  $S_T$  never enters.
2. **Spot repurchase plus cash settlement of the long.** Collect the loan, buy the ounce at  $S_T$ , and receive  $S_T - K$  on the long forward (gain when  $S_T$  is above  $K$ , pay when  $S_T$  is below). The combined cost to replace the ounce is

$$-S_T + (S_T - K) = -K.$$

The  $S_T$  terms cancel. You pay the delivery price  $K$  in net, exactly as under physical delivery. After the loan is collected, the same residual remains.

Line up the cash flows on the same timeline.

**Today ( $t = 0$ ) — same for every later ending of  $S_T$ :**

| Line                                          | Cash (USD) |
|-----------------------------------------------|------------|
| Sell one ounce in the spot market             | +4401.56   |
| Lend the proceeds at 4.1723%                  | −4401.56   |
| Long the forward at delivery price $K = 4400$ | 0          |
| <b>Net today</b>                              | <b>0</b>   |

You no longer hold the ounce, you have cash on deposit, and you are long the forward. No net cash has left your pocket yet.

**Maturity  $T$  — physical delivery under the long forward ( $S_T$  never appears):**

| Line                                                | Cash (USD)     |
|-----------------------------------------------------|----------------|
| Collect the loan repayment $S_0(1+r)^T$             | +4578.40       |
| Pay the delivery price $K$ ; receive the ounce back | −4400.00       |
| <b>Net at maturity <math>T</math></b>               | <b>+178.40</b> |

That is the cleanest picture of the reverse lock. The next example opens maturity  $T$  into a spot repurchase plus long settlement, so you can watch  $S_T$  appear and then cancel.

#### Same reverse cash-and-carry, three values of $S_T$

Keep the trade above: delivery price  $K = 4400$ , loan repayment  $\approx 4578.40$ . Today is unchanged — net cash at  $t = 0$  is still zero. At maturity  $T$ , collect the loan, buy the ounce at  $S_T$ , and settle the long forward for  $S_T - K$ .

**Case  $S_T > K$ .** Suppose gold finishes at  $S_T = 4700$ . Replacing the ounce in the spot market costs \$4,700 — painful after you sold at \$4,401.56. The long forward pays you back:

$$S_T - K = 4700 - 4400 = 300.$$

Net cost to replace the ounce:  $4700 - 300 = 4400$ . Collect the loan \$4,578.40; net  $\approx 178.40$ . The rally made the repurchase expensive and paid you on the long by the same amount.

**Case  $S_T = K$ .** Suppose gold finishes at the delivery price,  $S_T = 4400$ . The spot repurchase costs \$4,400 and the long forward settles at zero. After the loan, net  $\approx 178.40$ . When  $S_T = K$ , the forward is idle and the whole residual is just the loan repayment minus the delivery price  $K$ .

**Case  $S_T < K$ .** Suppose gold finishes at  $S_T = 4000$ . The spot repurchase is cheap at \$4,000, but the long forward is underwater:

$$S_T - K = 4000 - 4400 = -400.$$

Net cost to replace the ounce:  $4000 + 400 = 4400$ . Collect the loan; net  $\approx 178.40$  again. The fall made the repurchase cheap and cost you on the long by the same amount.

The same maturity- $T$  cash flows in one grid. Each column is one ending of  $S_T$ ; every column still nets to about \$178.40:

| Line at maturity $T$                 | $S_T > K$      | $S_T = K$      | $S_T < K$      |
|--------------------------------------|----------------|----------------|----------------|
| $S_T$                                | 4700.00        | 4400.00        | 4000.00        |
| Collect the loan repayment           | +4578.40       | +4578.40       | +4578.40       |
| Buy the ounce ( $-S_T$ )             | −4700.00       | −4400.00       | −4000.00       |
| Long settle ( $S_T - K$ )            | +300.00        | 0.00           | −400.00        |
| Subtotal to replace ounce ( $= -K$ ) | −4400.00       | −4400.00       | −4400.00       |
| <b>Net</b>                           | <b>+178.40</b> | <b>+178.40</b> | <b>+178.40</b> |

In every column, buying the ounce and settling the long costs the delivery price  $K = 4400$  — the same subtotal as paying  $K$  under physical delivery. Add the loan repayment ( $4578.40 + (-4400)$ ):

$$4578.40 - 4400 \approx 178.40.$$

That \$178.40 was fixed *today*, when the quoted forward sat below the financed cost of the ounce. Nothing that happens to  $S_T$  afterward opens or closes the gap.

#### Python — reverse lock vs three terminal spots

```
K = 4400.0
loan = 4578.40
for label, S_T in (
    ("above K", 4700.00),
    ("equal K", 4400.00),
    ("below K", 4000.00),
):
    buy = -S_T
    long_settle = S_T - K
    subtotal = buy + long_settle
    net = loan + subtotal
    print(label, buy, long_settle, subtotal, net)
```

**Key Insight.** Reverse cash-and-carry is not a bet on whether gold will rise or fall. Today ( $t = 0$ ) the cash flows net to zero. At maturity  $T$ , the long forward cancels the spot repurchase, whether  $S_T$  finishes above, at, or below the delivery price  $K$ . The locked profit is only the gap between the loan repayment and that delivery price  $K$ .

### 2.8.5 Storage and lease

Real gold is not a zero-storage, zero-income claim.

- **Storage.** If holding the metal costs money (vaulting, insurance), write  $U$  for the present value of storage over the life of the contract. The interest-only formula becomes  $F_0 = (S_0 + U)(1 + r)^T$  in the annual dialect (or  $F_0 = (S_0 + U)e^{rT}$  if  $r$  is continuous). Storage pushes the fair forward *up*.
- **Lease / income.** If holding gold earns a lease rate (lending the metal to a jeweller or a short seller), treat that income like a dividend yield  $q$ . Then  $F_0 = S_0(1 + r)^T / (1 + q)^T$  annually, or  $F_0 = S_0e^{(r-q)T}$  continuously. Lease income pushes the fair forward *down*.

The cash-and-carry illustration treated  $U = 0$  and  $q = 0$ , so the whole \$9.75 screen gap looked lockable. In a live metal market that residual is usually storage, lease, day-count, and the difference between an OTC forward story and an exchange future — a carry check, not automatic arbitrage. Even after those carry items are recognised, liquid markets rarely leave a textbook free lunch sitting on the screen.

### 2.8.6 Why live markets rarely offer a free lunch

By *screen* we mean the live spot and forward prices you see on a terminal.

On major gold, FX, and equity-index markets today, a gap that survives transaction costs, financing, and the true carry of the underlying is uncommon in quiet conditions. Dedicated capital watches the same screens; when a quote drifts outside the no-arbitrage band, cash-and-carry or reverse cash-and-carry tends to push it back before a casual trader can lock a risk-free profit.

That tightness is decades in the making, not a switch that flipped on one date. Before listed financial futures and electronic matching linked cash and forward markets tightly, mispricings could last longer and be harvested with slower tools. Stock-index futures (from 1982 on the major U.S. contracts) made cash-versus-futures arbitrage a visible desk business; through the 1990s and 2000s, electronic trading, thinner bid–ask spreads, and more balance-sheet dedicated to relative-value trades compressed the easy gaps on liquid underlyings.<sup>4</sup>

What changed is less the algebra than the market plumbing: continuous joint quoting of spot and forward or futures, cheap electronic execution, and many firms running the same carry check. The formula  $F_0 = S_0(1 + r)^T$  did not become false; the window in which a liquid quote could sit far from it *and* stay there shrank.

A screen residual is still not proof of free money. After 2008, covered interest parity in G10 FX — the FX cousin of cash-and-carry — often shows persistent cross-currency bases. Those gaps look like arbitrage until bank funding costs, capital, and balance-sheet limits are included; for constrained intermediaries they are not a risk-free lock.<sup>5</sup>

Where can something closer to an arbitrage still appear? Mostly away from the quiet, liquid core: less liquid OTC names, some emerging-market underlyings, stressed episodes when financing or borrow is rationed, and structures whose legs are hard to hedge or fund (delivery optionality, special collateral, credit or funding frictions). Those trades are usually relative-value books with risk, capital, and operational cost — not the classroom one-line lock on a COMEX gold future versus XAU.

**Key Insight.** Cash-and-carry when the quoted forward is above  $F_0$ ; reverse cash-and-carry when it is below. The lock is clean only under the stated carry assumptions. On a liquid metal or index screen, first subtract interest-only fair value; what remains is usually storage, lease, and instrument basis — and even that residual is rarely free money once real financing and balance-sheet costs are counted.

## 2.9 Which Financing Rate?

The fair forward  $F_0 = S_0(1 + r)^T$  treats  $r$  as the interest cost of borrowing cash to hold the asset until maturity  $T$ . So  $r$  must be a rate for *that horizon* — roughly six months for a six-month forward, roughly one year for a one-year forward.

<sup>4</sup>CME S&P 500 futures began in April 1982; CME Globex electronic matching launched in 1992; the electronically traded E-mini S&P 500 followed in 1997.

<sup>5</sup>See Du, W., Tepper, A., & Verdelhan, A. (2018). Deviations from covered interest rate parity. *Journal of Finance*, 73(3), 915–957.

### 2.9.1 Overnight SOFR is only one day

**SOFR** (Secured Overnight Financing Rate) is a USD interest rate for a loan that lasts *one business day*. It is published by the New York Fed from real Treasury-collateralised overnight loans. Because the loan is short and collateralised, SOFR is treated as near risk-free compared with unsecured bank borrowing.

One-day money is the wrong input for a multi-month forward. Using overnight SOFR alone in  $F_0 = S_0(1 + r)^T$  would be like pricing a year-long rent with last night's hotel rate: related, but not the same contract.

### 2.9.2 A SOFR OIS fixed locks a longer horizon

Markets quote longer financing through a *SOFR OIS* (overnight index swap). In plain terms, the OIS turns a strip of future overnight SOFR fixings into a single fixed rate for a stated period — six months, one year, and so on. The number plugged in as  $r$  is that *fixed* OIS rate for a tenor close to maturity  $T$ .

For the near one-year gold check ( $T \approx 0.96$ ), that rate is

$$r = 4.1723\%$$

from **US05FR1** Curncy — the *1-year* SOFR OIS fixed. A one-year-ish forward belongs on a one-year financing rate, not on overnight SOFR alone.

### 2.9.3 How the quote is written

These OIS fixed rates are usually quoted as annual rates on an Act/360 basis: interest scales with the actual number of calendar days in the period and a 360-day year,

$$\text{Interest} \approx \text{Principal} \times r \times (\text{days}/360).$$

They are not continuously compounded quotes. Putting the same printed  $r$  into  $e^{rT}$  is an idealised continuous dialect;  $S_0(1 + r)^T$  sits closer to the annual street quote. Neither formula rebuilds a full OIS day-count pricer — both are carry checks, not production pricing engines.

Older books often wrote “LIBOR” in this financing slot. USD panel LIBOR was an unsecured survey rate and was wound down in 2023. The modern USD input for the same role is SOFR / OIS.

**Key Insight.** Choose a financing rate whose length matches maturity  $T$ . Overnight SOFR is one day; a SOFR OIS fixed is a locked rate for a stated multi-month or multi-year tenor. The same quoted rate  $r$  produces two different fair forwards if it is plugged into  $e^{rT}$  or into  $(1 + r)^T$ . Those are two compounding conventions, not two different markets.

## 2.10 Forward Price Versus Value of a Forward

The *forward price*  $F_0$  and the *value* of a forward contract are easy to mix up. They are not the

same object.<sup>6</sup>

$F_0$  is a *delivery price*: the level of  $K$  that makes a *brand-new* forward for maturity  $T$  worth zero to enter today. Once you strike the deal, your contract keeps that locked delivery price  $K$ . The live forward price  $F_0$  can still move as spot and rates move.

The *value* of your existing forward is what that locked deal is worth *now* — how much someone would pay (or need to be paid) to take your side.

### 2.10.1 When the live forward price moves

Return to the non-dividend stock with continuous compounding:  $S_0 = 30$ ,  $r = 5\%$ ,  $T = 0.5$ , so a new forward would set

$$F_0 = 30e^{0.05 \times 0.5} \approx 30.76.$$

Suppose you go *long* at that level: delivery price  $K = 30.76$ . At inception the contract is worth zero — that is what  $K = F_0$  means.

Later the same day, imagine the live forward price for the same maturity  $T$  has moved to  $F_0 = 32.00$  (spot rose, or financing changed). A brand-new long would have to lock  $K = 32.00$ . Your old long still buys at 30.76. At maturity  $T$  you are better off than a new long by

$$32.00 - 30.76 = 1.24$$

per share. That future advantage has present value today. With continuous compounding,

$$V_{\text{long}} = e^{-rT}(F_0 - K) = e^{-0.05 \times 0.5}(32.00 - 30.76) \approx 1.21.$$

With annual compounding the same idea is

$$V_{\text{long}} = (F_0 - K)/(1 + r)^T.$$

The short's value is the opposite sign:  $V_{\text{short}} = -V_{\text{long}}$ .

#### Python — value of an existing long forward

```
import math
K = 30.76
F0 = 32.00
r = 0.05
T = 0.5
V_long = math.exp(-r * T) * (F0 - K)
print(f"V_long = {V_long}")
```

### 2.10.2 Forward price and value side by side

- **Forward price  $F_0$ :** the delivery price that makes a *new* forward worth zero.
- **Value of an existing forward:** the present value of being locked at an old delivery price  $K$  when the live forward price is now  $F_0$ . At inception  $K = F_0$  and value is zero; afterward, if  $F_0$  moves away from  $K$ , value is generally not zero.

<sup>6</sup>Hull, J. C. (2022). *Options, futures, and other derivatives* (11th ed., Global). Pearson. Practice Question 5.1.

**Key Insight.**  $F_0$  is a live delivery-price quote for new deals. An old forward's value is the discounted gap between that live  $F_0$  and the delivery price  $K$  already locked in the contract.

## 2.11 Chapter Summary and Formula Sheet

**Settlement P&L:**  $\text{P\&L} = \text{Quantity} \times (S_T - K) \times \text{Position}$   
 (Position = +1 long, -1 short)

**Fair forward (no income), continuous:**  $F_0 = S_0 e^{rT}$

**Fair forward (no income), annual:**  $F_0 = S_0(1 + r)^T$

**With storage PV  $U$  (annual):**  $F_0 = (S_0 + U)(1 + r)^T$

**With lease / income yield  $q$  (annual / continuous):**

$$F_0 = S_0(1 + r)^T / (1 + q)^T, \quad F_0 = S_0 e^{(r-q)T}$$

**Cash-and-carry:** quoted forward above  $F_0 \rightarrow$  buy spot, finance, short forward.

At maturity  $T$ :  $S_T + (K - S_T) = K$ ; locked profit =  $K - S_0(1 + r)^T$  (zero storage / income).

**Reverse cash-and-carry:** quoted forward below  $F_0 \rightarrow$  sell spot, lend, long forward.

At maturity  $T$ :  $-S_T + (S_T - K) = -K$ ; locked profit =  $S_0(1 + r)^T - K$ .

**Value of an existing long forward:**

$$V_{\text{long}} = e^{-rT}(F_0 - K) \text{ (continuous)}, \quad V_{\text{long}} = (F_0 - K)/(1 + r)^T \text{ (annual)}$$

$$V_{\text{short}} = -V_{\text{long}}$$

**$K$  vs  $F_0$ :** delivery price locked in an existing contract vs live delivery price that makes a *new* forward worth zero.

## 3 Futures Contracts

### 3.1 Why Trade Through an Exchange

Chapter 2 priced a *forward*: a private promise between two counterparties to exchange an asset for cash at a predetermined delivery price  $K$  on maturity date  $T$ . The economics were clean — long and short P&L, fair forward price  $F_0$ , cash-and-carry when the quote is wrong — but the contract itself stayed over the counter (OTC). Each side depends on the other to perform when maturity arrives.

This chapter keeps the same three ideas and changes the *venue*:

1. Identify the future exchange (what is bought or sold, how much, and when).
2. Value the position as the market moves — now through daily settlement, not only at maturity  $T$ .
3. If two routes deliver the same future cash, they must cost the same today (no free lunch). Calendar mispricings return with the same machinery as Chapter 2, now on exchange-traded contracts.

A *futures* contract is the exchange-traded cousin of the forward. The rest of the chapter builds standardisation and clearing, margin and marking to market, hedging with listed contracts, and a June–December crude calendar that re-derives cash-and-carry.

### 3.2 Futures Versus Forwards

A forward is bilateral. You and your counterparty negotiate size, maturity date  $T$ , and delivery price  $K$ . There is usually no daily cash until maturity (or until you unwind bilaterally). If the other side defaults when the contract is deep in their favour, you bear the loss.

A futures contract answers that problem with three design choices.

- **Standardisation.** The exchange fixes contract size, deliverable grade, tick size, and a menu of maturity months. Traders do not rebuild the legal terms for every deal; they trade the listed strip.
- **Central clearing.** Once the trade matches, the clearing house becomes the buyer to every seller and the seller to every buyer. Your counterparty risk is to the clearing system, not to the name on the other side of the screen.
- **Daily settlement.** Futures are *marked to market*: each day the clearing house credits or debits margin accounts for that day's price change. Gains and losses do not wait until maturity  $T$ .

#### Futures Contract

A standardised exchange-traded contract to buy or sell a specified quantity of an underlying

asset at a futures price that is marked to market daily. The clearing house stands between the long and the short; both post margin.

Long and short mean the same economic obligations as in a forward: the long is committed to *buy* (or settle as a buyer) and the short to *sell*. What changes is the path of cash before maturity  $T$  — variation margin moves every day — and the fact that almost all listed futures are closed out or cash-settled rather than held to physical delivery.

**Key Insight.** Forward: OTC, custom terms, performance risk to your counterparty, settlement concentrated at maturity  $T$ . Futures: standardised, cleared, both sides post margin, profits and losses settle day by day.

### 3.3 Clearing and Daily Settlement

When you enter a futures position you do not wire the full notional. You post an *initial margin* — collateral held by the clearing broker / clearing house against adverse moves. As the futures price moves, the clearing house *marks the position to market*:

- If the futures price rises, the long's margin account is credited and the short's is debited (by the contract size times the price change).
- If the futures price falls, the short is credited and the long is debited.

That daily cash transfer is *variation margin*. Unlike a typical forward, the exchange does not wait until maturity  $T$  to settle who won.

A second level, the *maintenance margin*, is the floor the account must stay above. If losses push the balance below maintenance, the trader receives a *margin call* and must top the account back up to the initial margin level (or close the position). Failure to meet the call leads to liquidation by the broker / exchange.

The notional at risk is larger than the margin. Contract value equals contract size times the futures price. A one-cent move on a 5,000-ounce silver contract is already

$$5000 \times 0.01 = 50$$

dollars of variation margin — fifty dollars of cash, long or short, for a single penny.

### 3.4 Worked Example: Short Silver Futures Margin

A trader enters a *short* COMEX silver futures at the market. Contract size is 5,000 ounces. How far can the futures price rise before a margin call? What happens if the call is not met?<sup>7</sup>

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<sup>7</sup>Same margin arithmetic as Hull, J. C. (2022). *Options, futures, and other derivatives* (11th ed., Global). Pearson. Practice Question 2.1; levels and SPAN margins from 15 September 2026.

**Silver short (15 September 2026)**

Short one COMEX silver futures (SIU6, 5,000 oz) at last \$63.030/oz. Speculator initial margin \$34,932; maintenance margin \$31,757.

Contract value at entry:

$$5000 \times 63.030 = 315150$$

Only \$34,932 sits as initial margin against that \$315,150 notional — roughly one-ninth of the contract value, not the full silver bill.

**Python — contract value and one-cent move**

```
size = 5000
futures_price = 63.030
contract_value = size * futures_price
one_cent_pnl = size * 0.01
print(f"contract_value = {contract_value}")
print(f"one_cent_pnl = {one_cent_pnl}")
```

The short loses when the futures price *rises*. The cushion above maintenance is

$$34932 - 31757 = 3175$$

A margin call hits once cumulative losses reach that cushion. Dividing by contract size converts the dollar cushion into a price move:

$$\Delta = \frac{34932 - 31757}{5000} = \frac{3175}{5000} = 0.635$$

So a rise of \$0.635 per ounce triggers the call. From the entry last \$63.030, that is a futures price of

$$63.030 + 0.635 = 63.665$$

**Python — margin-call price move (short)**

```
size = 5000
initial = 34_932
maintenance = 31_757
entry = 63.030

cushion = initial - maintenance
delta = cushion / size
call_price = entry + delta # short: call when futures rises

print(f"cushion = {cushion}")
print(f"delta = {delta}")
print(f"call_price = {call_price}")
```

In general, for a short futures position,

$$\Delta_{\text{call}} = \frac{\text{initial margin} - \text{maintenance margin}}{\text{contract size}}$$

and the call level is the entry futures price plus  $\Delta_{\text{call}}$ . A long would face the same-sized  $\Delta_{\text{call}}$  on a *fall*.

If the margin call is not met, the clearing house closes out the position. It does not wait until the delivery month.

**Key Insight.** Margin is a small cash buffer against daily mark-to-market losses, not payment of the full contract value. For a short, the call arrives after a rise of (initial – maintenance)/ size; miss the call and the position is liquidated.

### 3.5 Hedging with Futures

A futures hedge offsets a price risk that already exists in the physical business — the same economic motive as the forward hedges in Chapter 2, now on a listed contract with daily margin.

- **Seller of the good** (producer, farmer, miner): already long the physical. Take a *short* futures to protect against a price fall.
- **Buyer of the good** (manufacturer, airline, jeweller): already short the physical in economic terms. Take a *long* futures to protect against a price rise.

What you lose on one side, you gain on the other when the hedge is perfect. The locked outcome is a known selling or purchase price (up to basis). The cost of that certainty is giving up favourable moves: the short hedge truncates upside in the cash market; the long hedge truncates the benefit of a cheaper spot.

*Basis risk* remains. The local cash price need not move one-for-one with the futures (grade, location, timing). Hedging reduces exposure; it does not delete every residual.

### 3.6 Worked Example: Cattle Farmer Hedge

A cattle farmer expects to have 120,000 pounds of live cattle to sell in about three months. How many CME live-cattle futures should the farmer short, and what are the trade-offs?<sup>8</sup>

#### Short live-cattle futures (15 September 2026)

Exposure 120,000 lb of finished cattle to sell in about three months. CME live-cattle contract size on LEZ6: 44,000 lb. Nearby futures last: 174.00 ¢/lb.

Number of contracts:

$$120000/44000 \approx 2.73$$

<sup>8</sup>Same hedge logic as Hull, J. C. (2022). *Options, futures, and other derivatives* (11th ed., Global). Pearson. Practice Question 2.19; contract size from CME live cattle on 15 September 2026 (LEZ6).

Round to the nearest whole contract: *short three* live-cattle futures. The farmer is long the herd, so the hedge sells the listed futures against the future physical sale. Three contracts cover (3 times 44000 = 132000) lb, about 12,000 lb more than the planned sale — a small over-hedge from rounding.

#### Python — hedge contract count

```
exposure_lb = 120_000
contract_size_lb = 44_000
n_exact = exposure_lb / contract_size_lb
n_contracts = round(n_exact) # short 3
residual_lb = n_contracts * contract_size_lb - exposure_lb
print(f"n_exact = {n_exact}")
print(f"n_contracts = {n_contracts}")
print(f"residual_lb = {residual_lb}")
```

**Pros.** The short locks a selling price close to the futures quote (up to basis). It cuts exposure to a fall in cattle prices over the next three months. The listed contract is standardised and liquid compared with negotiating a one-off forward with a single packer.

**Cons.** Basis risk: local cash cattle need not track the CME contract exactly. The farmer gives up the rally — if cattle prices rise, futures losses offset herd gains. Variation margin can demand cash *before* the animals are sold. Live production is rarely an exact multiple of the contract size — here the 44,000 lb unit leaves a few thousand pounds of mismatch even after rounding to three.

## 3.7 Calendar Cash-and-Carry

Chapter 2 compared a quoted forward to the financed cost of holding the spot asset. The same idea applies across two futures on the same commodity: treat the *nearby* contract as the synthetic spot, finance it to the *deferred* maturity, and compare that fair level to the deferred futures price. If the deferred contract is too rich and storage is free, buy nearby, sell deferred, take delivery, hold, and deliver — cash-and-carry. If the deferred is too cheap, that trade locks a loss instead; the residual is a carry check, not automatic arbitrage.<sup>9</sup>

#### June–December WTI (15 September 2026)

June 2027 futures (CLM7) last \$77.970/bbl; December 2027 (CLZ7) last \$72.710/bbl. Six-month financing  $r = 4.5856\%$  (US05FR6); storage treated as zero for the  $r$ -only check;  $T = 0.5$  year from June to December.

Treat the June futures as the price of oil in June. The fair December futures relative to that June level, with simple accrual (OIS / money-market style), is

$$F_0 = 77.970(1 + 0.045856 \times 0.5) \approx 79.758.$$

<sup>9</sup>Same calendar arithmetic as Hull, J. C. (2022). *Options, futures, and other derivatives* (11th ed., Global). Pearson. Practice Question 2.25; levels from NYMEX WTI and SOFR OIS on 15 September 2026.

December last is \$72.710 — about \$7 *below* the interest-only fair level, not above it.

#### Python — fair December vs quoted

```
june = 77.970
dec_quoted = 72.710
r = 0.045856
T = 0.5
F0 = june * (1 + r * T)
residual = dec_quoted - F0
print(f"F0 = {F0}")
print(f"residual = {residual}")
```

Run the cash-and-carry trade anyway — long June, short December — to see what the same cash-flow machine produces when the deferred contract is cheap.

**Today ( $t = 0$ ).** Futures cost nothing up front to enter:

| Line at $t = 0$          | Cash / bbl |
|--------------------------|------------|
| Long June at 77.970      | \$0        |
| Short December at 72.710 | \$0        |
| <b>Net today</b>         | <b>0</b>   |

**In June.** Take delivery on the long, finance the barrel, store for free:

| Line in June               | Cash / bbl |
|----------------------------|------------|
| Pay June; take oil         | \$−77.970  |
| Borrow \$77.970 at 4.5856% | \$+77.970  |
| Store (free)               | \$0        |
| <b>Net in June</b>         | <b>0</b>   |

The loan grows to  $(77.970 (1 + 0.045856 \text{ times } 0.5))$  approx 79.758) by December.

**In December — physical delivery under the short.** Deliver the barrel, receive 72.710, repay the loan:

| Line in December           | Cash / bbl            |
|----------------------------|-----------------------|
| Deliver; receive 72.710    | \$+72.710             |
| Repay loan (approx 79.758) | \$−79.758             |
| <b>Net in December</b>     | <b>(approx −7.05)</b> |

December's oil price never appears under physical delivery. The lock is a *loss* of about \$7/bbl: the short December futures pays too little relative to the financed June barrel. Storage, convenience yield, and physical tightness sit inside that residual; an  $r$ -only screen is a carry check, not a free lunch.

### 3.7.1 Three December spots, one locked residual

The same residual survives if you sell the oil in the spot market at  $S_{\text{Dec}}$  and cash-settle the short December futures. Spot sale plus short settlement always rebuilds 72.710:

$$S_{\text{Dec}} + (72.710 - S_{\text{Dec}}) = 72.710.$$

After repaying the (approx \$79.758) loan, the net is still about \$−7.05 — whether December oil finishes above, at, or below 72.710.

| December           | $S > 72.71$ | $S = 72.71$ | $S < 72.71$ |
|--------------------|-------------|-------------|-------------|
| $S_{\text{Dec}}$   | 80.00       | 72.71       | 65.00       |
| Sell oil           | \$+80.00    | \$+72.71    | \$+65.00    |
| Short settle       | \$−7.29     | \$0         | \$+7.71     |
| Subtotal (= 72.71) | \$+72.71    | \$+72.71    | \$+72.71    |
| Repay loan         | \$−79.76    | \$−79.76    | \$−79.76    |
| Net                | \$−7.05     | \$−7.05     | \$−7.05     |

#### Python — three December spots, same residual

```
loan = 77.970 * (1 + 0.045856 * 0.5)
dec_futures = 72.710
for S in (80.0, 72.710, 65.0):
    sell = S
    short_settle = dec_futures - S
    net = sell + short_settle - loan
    print(f"S={S}: subtotal={sell + short_settle}, net={net}")
```

## 3.8 Contango and Backwardation

Plot futures prices against maturity. When deferred contracts sit above nearby ones, the curve slopes up — *contango*, the usual shape when financing and storage dominate. When nearby sits above deferred, the curve slopes down — *backwardation*.

The NYMEX WTI strip on 24 September 2026 is firmly backwardated: the front month (CLX6, November 2026) last at \$95.13/bbl, while contracts two years out sit near \$69–70. June 2027 (CLM7) and December 2027 (CLZ7) — the maturities used in the calendar worked example — last at \$79.37 and \$73.90 on this tape. Near-term oil is scarce relative to later months; the market pays up for barrels *now*.

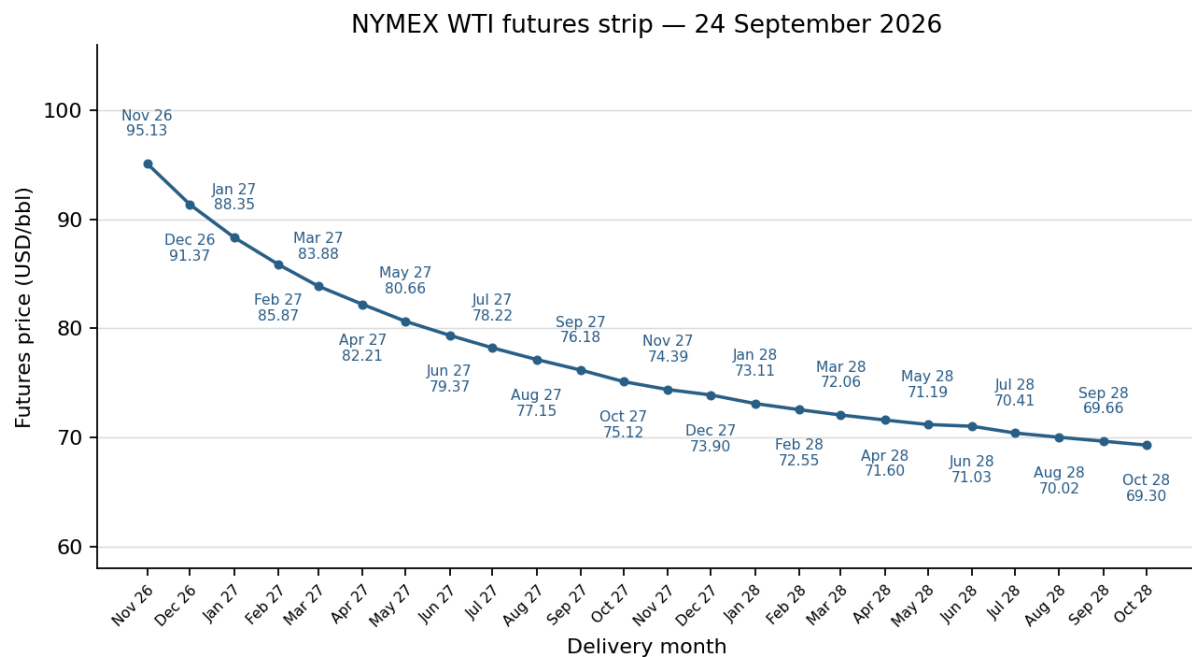


Figure 2: NYMEX WTI futures strip, lasts on 24 September 2026 (CL1 through CL24). The curve slopes down — backwardation — from about \$95/bbl in the front to about \$69 two years out. COMEX gold on the same date is the opposite shape. The listed strip through October 2028 rises with maturity: October 2026 last at \$4,268.20/oz, climbing to \$4,716.90 by October 2028. That upward slope is contango — the financing cost of carrying gold, in the same spirit as the Chapter 1–2 cash-and-carry checks.

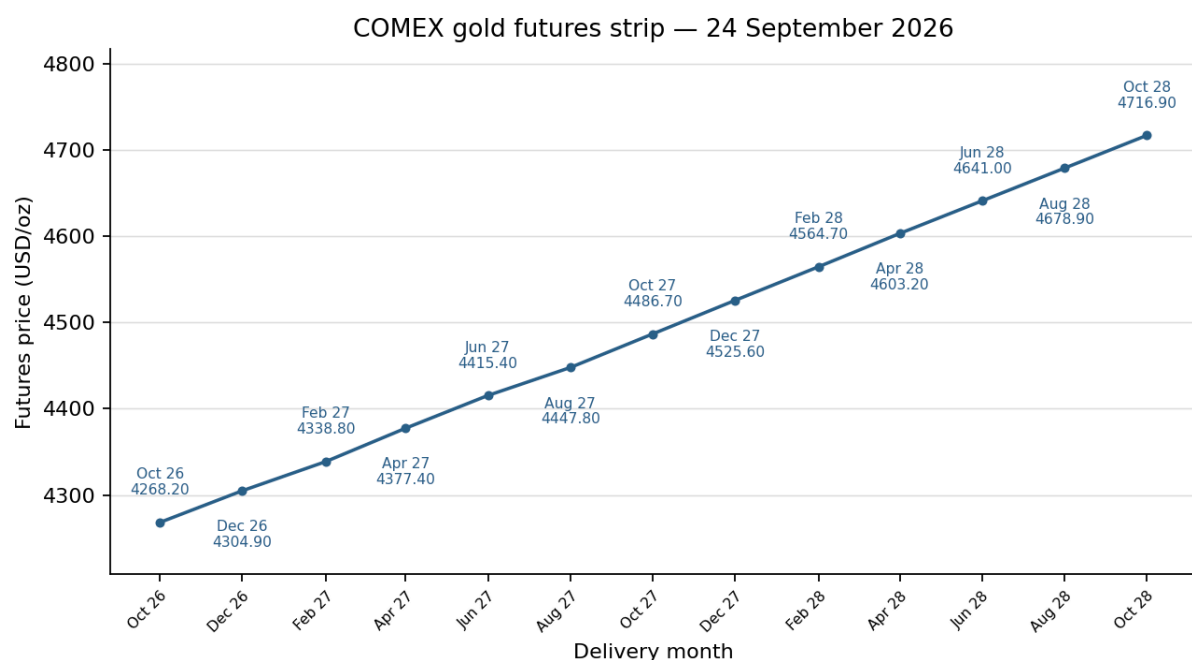


Figure 3: COMEX gold futures strip through October 2028, lasts on 24 September 2026. The curve slopes up — contango — from about \$4,268/oz in the front to about \$4,717 two years out. Gold lists even delivery months, so the ticks are the contracts that trade, not every calendar month.

### 3.9 Chapter Summary and Formula Sheet

**Futures vs forward:** standardised, cleared, daily mark-to-market and margin; forward is OTC with settlement concentrated at maturity  $T$ .

**Variation margin (one day):**  $\Delta \text{ cash} = \text{contract size} \times \Delta \text{ futures price}$   
(long credited when the futures price rises; short when it falls)

**Margin call (short):**  $\Delta_{\text{call}} = (\text{initial margin} - \text{maintenance margin}) / \text{contract size}$   
Call level = entry futures price  $+\Delta_{\text{call}}$ . Long: same  $\Delta_{\text{call}}$  on a fall.  
Miss the call  $\rightarrow$  position liquidated.

**Hedge size:**  $n \approx \text{exposure} / \text{contract size}$  (round to a whole number of contracts)  
Producer / seller  $\rightarrow$  short futures; buyer / consumer  $\rightarrow$  long futures.  
Basis risk and variation-margin cash-flow remain.

**Calendar fair deferred (simple, zero storage):**  $F_0 = F_{\text{nearby}}(1 + rT)$   
Residual = quoted deferred  $-F_0$ . Positive  $\rightarrow$  deferred rich (cash-and-carry candidate);  
negative  $\rightarrow$  deferred cheap (carry check, not an automatic free lunch).

**Contango:** deferred above nearby (upward strip). **Backwardation:** nearby above deferred (downward strip).